nature portfolio

Peer Review File

AI-driven fusion of multimodal data for Alzheimer's disease biomarker assessment

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This manuscript has been previously reviewed at another journal. This document only contains information relating to versions considered at Nature Communications.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
None

(Remarks on code availability)
N/A

Reviewer #4

(Remarks to the Author)

The authors have significantly improved methodological clarity, including better explanation of the loss function, data masking, and spatial validation. They also respond reasonably to concerns about data overlap and PET harmonization. These steps reflect a solid computational effort.

However, several important concerns remain:

It is unclear whether the model was retrained after ComBat harmonization or deduplication, or whether these steps affected the sample size or performance. This should be clarified.

The model’s dependence on CSF biomarkers for strong performance undermines its claimed value as a practical, scalable alternative to PET.

The authors offer weak justification for building a new predictive model on top of plasma AD biomarkers, which are already approved and non-invasive. If plasma assays are accurate and accessible, the value of this added modeling layer needs a clearer rationale.

The authors critique plasma biomarkers for being influenced by non-neurological factors, yet many non-neurological variables are included in their own model (e.g., labs, medications, comorbidities), and no sensitivity analysis has been done to assess bias.

There are still unaddressed elements of circularity, especially in reported associations between model predictions and cognitive test scores (e.g., CDR, FAQ), some of which appear to have been used as inputs.

The shift in framing toward tau PET staging is interesting but premature; there is no validated clinical framework for how tau stages should influence patient care.

Finally, there is no assessment of performance in common alternative or mixed dementias (e.g., NPH, DLB, PSP, LATE, SD), limiting generalizability. This is a common limitation of the datasets used to train the model.

Recommendation:
This work represents a strong methodological contribution in multimodal model development, suitable for a journal focused on AI methodology. However, the clinical impact remains limited, and key issues around generalizability, robustness, and conceptual justification are unresolved.

(Remarks on code availability)

Reviewer #5

(Remarks to the Author)
The authors have thoroughly addressed the comments raised by all reviewers in the first round, and I commend them for their detailed and constructive responses. Overall, the manuscript has been substantially improved and clarified based on the previous feedback.

In particular, I appreciate that the authors carefully addressed the concerns regarding the dependency on CSF biomarkers. The authors convincingly demonstrated that their model maintains robust performance even in the absence of CSF data, which increases the applicability of their approach in settings where CSF is not routinely available. The clarification of the model's flexibility, to integrate high-fidelity biomarkers when available but still function with limited data, is well presented.

Furthermore, the authors have appropriately adapted the framing of their study towards disease staging rather than solely as a diagnostic tool. The additional analyses with the composite amyloid-tau (AT) score, as well as the differentiation of AD stages based on predicted regional tau distributions, are a valuable extension and strengthen the translational relevance of the work.
Regarding methodological transparency, I appreciate the authors' detailed revisions following Reviewer #3's concerns. The updated description of the multi-label classification framework, the use of focal loss, label masking strategy, and the handling of missing data now provide the necessary clarity. The steps taken to avoid data leakage across cohorts and the harmonization of PET data using ComBat are appropriate and well justified.

Moreover, the authors addressed limitations in the data structure and validation strategy transparently. The updated spatial analysis with spatial permutation testing and the shift to reporting AMI (Adjusted Mutual Information) rather than ARI further improve the robustness of the findings.

In summary, I believe the authors have carefully and convincingly addressed the reviewers' concerns and improved the manuscript substantially. The revisions enhance the clarity, methodological rigor, and clinical relevance of the study. While I have no major concerns remaining, I do have a minor comment for the authors.

In the rebuttal, and similarly in manuscript, the authors state: “In the future, when more data becomes available, we plan to incorporate fully quantitative outputs for Aβ and τ.” I would appreciate if the authors could further elaborate on this point. Specifically, how much additional data would be required to enable robust quantitative predictions? Given that the present study already combines several large and well-established cohorts, it would be helpful to better understand how realistic this plan is in the context of currently available or foreseeable datasets.

As a critical remark, I do not see that the current approach is really ready to change pathways since blood-based biomarkers alone facilitate AUCs between 0.93 and 0.96 to predict AD neuropathology (i.e. https://www.nature.com/articles/s41591-025-03622-w). The updated AUCs of 0.79-0.84 of the proposed apprach look rather low in this comparison. However, I acknowledge the authors idea of integrating blood-based biomarkers in their concept in future.

(Remarks on code availability)
i only checked the availability and found the code online. i am not an expert in coding, so i cannot judge if it is running properly

Reviewer #6

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)

Version 2:

Reviewer comments:

Reviewer #4

(Remarks to the Author)
If harmonization and deduplication steps did not matter for the study modeling/conclusions, then that is an important point to show quantitatively in the manuscript.

“Ptau217 remains an open issue and more research is needed before clinical implementation of specific plasma assays” These are widely available for clinical use via CLIA certified laboratories, and one has been FDA approved. Software relying on complex multi-model inputs with unsolved problems in real-world clinical workflows (e.g., combining cognitive assessments, MRI, laboratory tests, co-morbidities and medication data) in a resource poor setting that has a challenging time accurately reporting and organizing this data without also adding an AI modeling layer seems unrealistic, unnecessary, and certainly unproven by the current paper.

“Our study has a few limitations despite its strengths in scale, multimodal integration, and validation approach. Our model was developed and validated on seven distinct cohorts; however, its generalizability across diverse populations and clinical settings remains to be determined, as the dataset was predominantly composed of White participants. Importantly, due to the lack of non-AD and mixed dementia cases in our datasets, the generalizability of our findings to these important clinical phenotypes remains to be evaluated.” This is a good addition, but it should be clear that in resource poor routine clinical settings, non-AD causes of cognitive impairment are quite common making the claims of clinical utility of the current model that is ignorant of this fact overblown.

In general, the clinical applicability and impact of this work is highly speculative, but the manuscript does not read that way. It reads like a major clinical advance, but it is more of a technical proof of potential for multimodal data integration without a sound clinical rational at the moment.

(Remarks on code availability)

Version 3:

Reviewer comments:

Reviewer #4

(Remarks to the Author)
I appreciate the authors’ responsiveness. The methodological advance is clear and well-executed. However, I remain skeptical about its utility for clinical trial stratification or clinical practice. I have no further comments.

(Remarks on code availability)

Reviewers' comments:

Reviewer #1:

Comment R1C1: The authors have considerably revised and improved their original manuscript and have addressed my previous concerns. I am happy to accept the paper.

Response R1C1: We sincerely thank the reviewer for taking the time to re-review our revised manuscript. We especially appreciate their continued engagement as the sole reviewer in this round and are grateful for the supportive and constructive feedback. We are pleased that the revised manuscript meets the reviewer’s expectations.

Reviewer #4:

<u>Comment R4C1</u>: The authors have significantly improved methodological clarity, including better explanation of the loss function, data masking, and spatial validation. They also respond reasonably to concerns about data overlap and PET harmonization. These steps reflect a solid computational effort. However, several important concerns remain:

<u>Response R4C1</u>: We sincerely appreciate the reviewer's feedback on our revised manuscript.

<u>Comment R4C2</u>: It is unclear whether the model was retrained after ComBat harmonization or deduplication, or whether these steps affected the sample size or performance. This should be clarified.

<u>Response R4C2</u>: The model was trained only once following ComBat harmonization. To clarify, in response to reviewer feedback in the previous round of revisions, we implemented ComBat to address concerns regarding tau PET data harmonization and subsequently trained the model on the harmonized dataset. This was the only retraining step conducted. Neither ComBat harmonization nor deduplication led to any change in sample size or performance. We have updated the *PET positivity thresholding and tau profiling* subsection of our methods to clarify this point.

> A Gaussian mixture model (GMM) with two components was run on the ComBat-harmonized tau PET SUVr data from the training set and tau PET positivity was defined as SUVr values greater than 1.37. In addition to the meta-temporal ROI, we also defined tau ROIs associated with various AD stages and subtypes: medial temporal, lateral temporal, medial parietal, lateral parietal, frontal and occipital[17,18]. GMM analyses on the harmonized tau PET data set the positivity thresholds at 1.32, 1.33, 1.38, 1.29, 1.30 and 1.23, respectively.

<u>Comment R4C3</u>: The model's dependence on CSF biomarkers for strong performance undermines its claimed value as a practical, scalable alternative to PET.

<u>Response R4C3</u>: We would like to clarify that CSF biomarkers were included in initial model iterations (i.e., our first submission) to establish a performance benchmark, not as a requirement for practical deployment. In response to earlier concerns about the feasibility of CSF collection, we specifically evaluated the model <u>without</u> this input (i.e., in our earlier, revised submission). As expected, performance decreased modestly, but the model continued to demonstrate clinically meaningful accuracy using only non-invasive inputs such as cognitive assessments, MRI, laboratory tests, and medication data. This result is significant because it highlights the model's adaptability: it performs robustly in settings where CSF is unavailable, while also benefiting from enhanced accuracy when high-fidelity biomarkers are accessible. Rather than undermining its value, the inclusion and subsequent removal of CSF demonstrate a practical, tiered deployment strategy. Our approach mirrors real-world care environments, i.e., from well-resourced academic centers to community clinics, and supports informed decision-making across this spectrum. We believe this flexible design enhances the model's translational relevance and

represents a step forward in bridging heterogeneous, multi-modal data from various cohorts with everyday clinical workflows using advanced AI methods.

Comment R4C4: The authors offer weak justification for building a new predictive model on top of plasma AD biomarkers, which are already approved and non-invasive. If plasma assays are accurate and accessible, the value of this added modeling layer needs a clearer rationale.

Response R4C4: While plasma AD biomarkers represent an advancement due to their non-invasiveness and scalability, we respectfully argue that their standalone use is not sufficient for individualized decision-making. Our model does not aim to replace plasma assays, but rather to contextualize and augment them using routine clinical data, thereby improving reliability and reducing false positives or negatives in real-world populations. This layered modeling approach aims to better account for individual variability, enhance robustness, and potentially guide interpretation in patient populations where plasma measures alone may be noisy or misleading. Moreover, our findings demonstrate that even in the absence of plasma biomarkers, models using routinely collected neurology work-up data can achieve robust performance. This is especially valuable in resource-limited settings where blood-based biomarker testing is not yet widely available. Finally, variability among plasma assays regarding brain-specific versus peripheral p-tau217 remains an open issue and more research is needed before clinical implementation of specific plasma assays. As such, our model relies on features that are already well-integrated in clinical workflows, while offering the flexibility to seamlessly integrate newly validated biomarkers when those become more widely available. We believe this justifies our modeling approach as complementary but not redundant to existing plasma assays, providing an interpretable and accessible framework for early detection and triage in ADRD.

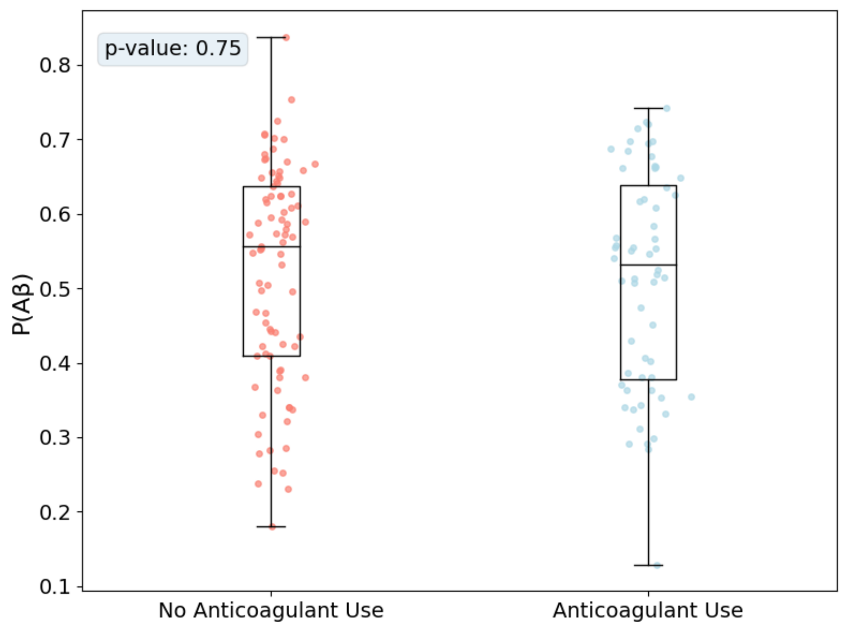
Comment R4C5: The authors critique plasma biomarkers for being influenced by non-neurological factors, yet many non-neurological variables are included in their own model (e.g., labs, medications, comorbidities), and no sensitivity analysis has been done to assess bias.

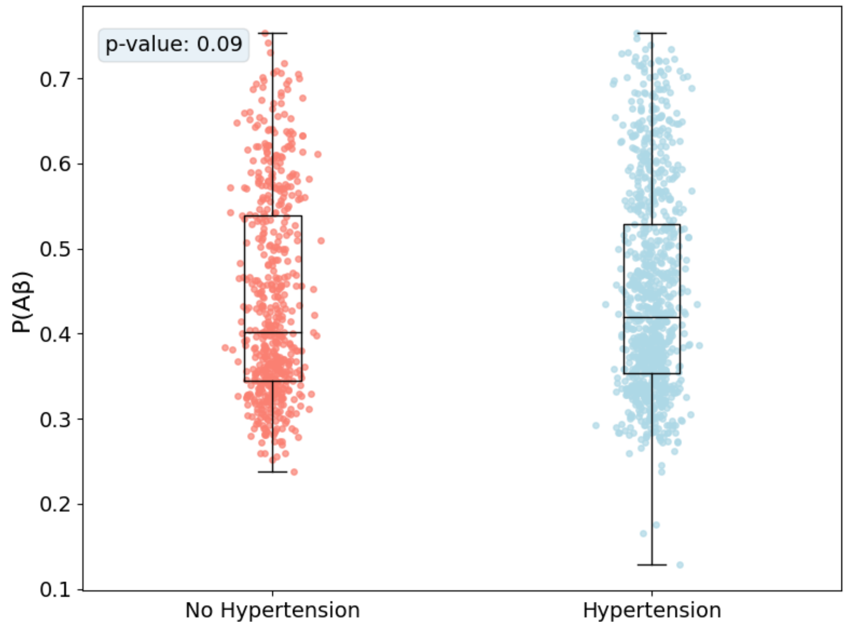
Response R4C5: Our note on plasma biomarkers being influenced by non-neurological factors was not intended to suggest that including such variables in predictive models is inherently problematic. Rather, we were highlighting how these biomarkers are often interpreted in isolation as specific proxies for AD pathology, despite known sensitivity to systemic factors such as renal function, inflammation, and medication use. This narrow interpretation is what we believe undermines their standalone specificity. In contrast, our model is designed to emulate the integrative reasoning of a neurologist. It does not prioritize any single input but instead synthesizes multimodal information including labs, medications, comorbidities, imaging, and cognitive assessments, to form a holistic diagnostic prediction. The architecture supports flexible, context-aware inference even in the presence of missing or noisy data.

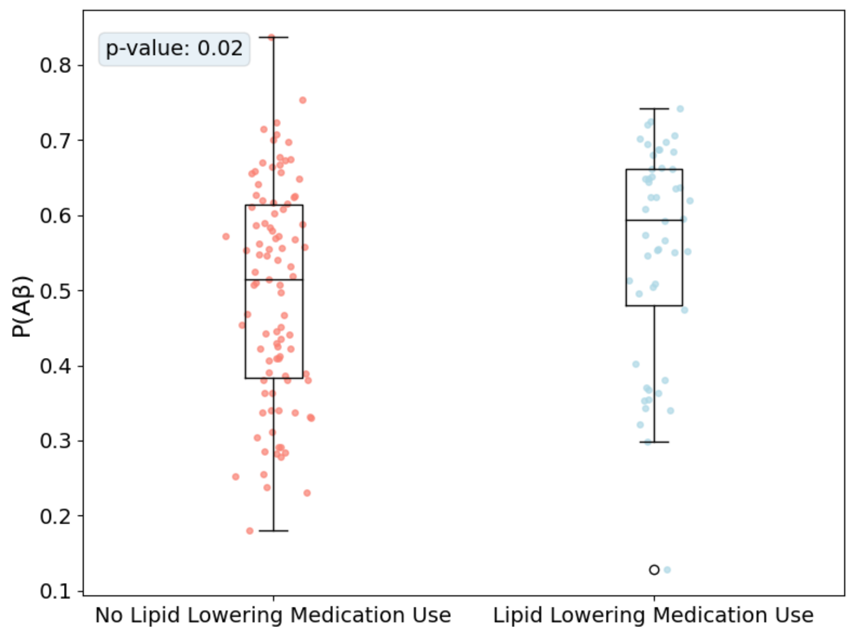
We conducted three subgroup sensitivity analyses using non-neurological variables for illustration purposes: current use of anticoagulant/antiplatelet medications (N=145), presence of hypertension (N=1500), and current use of lipid-lowering medication (N=145). We performed a two-sided Mann-Whitney test to evaluate whether the difference in median Aβ probability was statistically significant between groups.

As shown below, model-predicted probabilities of amyloid PET positivity were not significantly different across the first two subgroups (p -value= 0.75, p-value = 0.09, respectively). . We did, however, find a significant difference in model probabilities of amyloid positivity between cases with reported use of lipid-lowering medications and those without a reported use of these medications (U=2955, p-value = 0.02). These results are not presented to claim directional effects of any one factor but simply to illustrate that our model can operate flexibly across a range of input scenarios, including when certain non-neurological variables are present, absent, or variable in influence. Taken together, these results suggest that the model's outputs are not systematically biased by these non-neurological variables. Importantly, the model is designed to integrate all available data, including non-neurological inputs, as part of a holistic diagnostic inference, rather than relying on any single variable or modality in isolation.

We would like to emphasize that these subgroup analyses are not central to our main findings but are provided here to demonstrate robustness and flexibility of our multimodal framework. While both plasma biomarkers and our model may include signals from non-neurological sources, the key distinction is in how that information is integrated. Our model interprets data within a broader diagnostic context, avoiding overreliance on any one modality. Our goal is not to critique biomarker-based approaches per se, but to highlight the value of multimodal reasoning in reducing misinterpretation and bias.







<u>Comment R4C6</u>: There are still unaddressed elements of circularity, especially in reported associations between model predictions and cognitive test scores (e.g., CDR, FAQ), some of which appear to have been used as inputs.

<u>Response R4C6</u>: In the previous round of revision, we removed all analyses comparing model predictions with cognitive test scores. In the current version, we present model predicted probabilities in relation to Centiloid and tau PET SUVr values in Figure 3a, 3b and have colored the points by ADAS-13 scores. ADAS-13 was not used as input and is merely shown in this figure for visualization purposes to increase the clinical interpretability of our model predictions. Additionally, we would like to note that our model is not being validated against CDR and FAQ, as those were indeed used as inputs.

Comment R4C7: The shift in framing toward tau PET staging is interesting but premature; there is no validated clinical framework for how tau stages should influence patient care.

Response R4C7: We appreciate the reviewer's perspective regarding the evolving role of tau PET staging in clinical practice. While we agree that tau PET staging is not yet fully integrated as a validated tool for routine clinical decision-making, we believe that engaging with its potential relevance is timely and necessary, particularly considering recent advances in biomarker-driven frameworks and updated consensus guidelines.

Recent revisions to the AD diagnostic and staging criteria by the Alzheimer's Association explicitly recognize the prognostic value of Core 2 biomarkers, including tau PET. These criteria highlight that tau PET, when integrated with clinical data, can enhance diagnostic confidence and provide critical insights into disease progression. In fact, in the previous round of revisions, one reviewer constructively suggested framing our work in terms of tau staging, and we appreciated this recommendation. Accordingly, we adopted this framing to explore how AI models can support emerging biomarker frameworks. Recent studies, including those cited in our manuscript, demonstrate that tau PET staging is strongly associated with disease severity, symptom progression, and even treatment responsiveness, making it a compelling candidate for future stratification frameworks. Our intention is not to suggest that tau staging is already a validated clinical decision-making tool, but rather to illustrate how AI systems can help bridge the gap between emerging biomarker frameworks and real-world clinical implementation. By directly modeling tau PET stages alongside other biomarker and clinical modalities, our work provides a proof of concept for integrating heterogeneous data to comprehensively assess AD. This approach represents a step toward building systems that can aid in diagnosis and staging, leveraging clinically accessible data without drastically sacrificing biological precision.

Comment R4C8: Finally, there is no assessment of performance in common alternative or mixed dementias (e.g., NPH, DLB, PSP, LATE, SD), limiting generalizability. This is a common limitation of the datasets used to train the model.

Response R4C8: We agree with the reviewer that the limited representation of alternative or mixed dementias is a known and important limitation, not only of our study but of most widely used datasets in the ADRD field. We have acknowledged this limitation in the revised manuscript. Importantly, we recognize the critical need to expand model validation across broader phenotypes and have begun addressing this in our ongoing work. In fact, our group recently published a study in *Nature Medicine* (https://www.nature.com/articles/s41591-024-03118-z) where we evaluated the performance of a related model across several non-AD dementia types. We are actively extending this line of research to include additional cohorts with diverse neurodegenerative conditions, and we view this as an essential next step to establish the generalizability and clinical utility of AI-based tools in real-world settings.

Our study has a few limitations despite its strengths in scale, multimodal integration, and validation approach. Our model was developed and validated on seven distinct cohorts; however, its generalizability across diverse populations and clinical settings remains to be determined, as the dataset was predominantly composed of White participants. Importantly, due to the lack of non-AD and mixed dementia cases in our datasets, the generalizability of our findings to these important clinical phenotypes remains to be evaluated.

Comment R4C9: This work represents a strong methodological contribution in multimodal model development, suitable for a journal focused on AI methodology. However, the clinical impact remains limited, and key issues around generalizability, robustness, and conceptual justification are unresolved.

Response R4C9: We thank the reviewer for recognizing the methodological strength of our approach. We believe our work also demonstrates clinical relevance by addressing key translational barriers and offering a flexible framework that aligns with real-world diagnostic workflows. One of the core motivations of this work is to demonstrate that routinely collected multimodal data from standard neurology workups such as clinical history, cognitive tests, labs, medications, imaging, and comorbidities can be effectively leveraged by AI to infer biomarker status without requiring invasive or expensive procedures. This is especially relevant in real-world settings where access to PET or CSF may be limited. Furthermore, we discussed the trade-offs between model complexity, data availability, and clinical feasibility, and we show that our model retains clinically meaningful performance even in the absence of high-fidelity biomarkers. We believe this flexibility enhances the translational relevance of our work and offers a practical pathway for expanding access to precision diagnostics. To support generalizability, we also harmonized imaging data, avoided cohort leakage, and tested across multiple datasets. All steps were detailed in the revised manuscript. We view our work as a bridge between cutting-edge multimodal AI methods and clinical applicability and believe that this translational angle meaningfully advances the field.

Reviewer #5 (Remarks to the Author):

Comment R5C1: The authors have thoroughly addressed the comments raised by all reviewers in the first round, and I commend them for their detailed and constructive responses. Overall, the manuscript has been substantially improved and clarified based on the previous feedback. In particular, I appreciate that the authors carefully addressed the concerns regarding the dependency on CSF biomarkers. The authors convincingly demonstrated that their model maintains robust performance even in the absence of CSF data, which increases the applicability of their approach in settings where CSF is not routinely available. The clarification of the model's flexibility, to integrate high-fidelity biomarkers when available but still function with limited data, is well presented. Furthermore, the authors have appropriately adapted the framing of their study towards disease staging rather than solely as a diagnostic tool. The additional analyses with the composite amyloid-tau (AT) score, as well as the differentiation of AD stages based on predicted regional tau distributions, are a valuable extension and strengthen the translational relevance of the work. Regarding methodological transparency, I appreciate the authors' detailed revisions following Reviewer #3's concerns. The updated description of the multi-label classification framework, the use of focal loss, label masking strategy, and the handling of missing data now provide the necessary clarity. The steps taken to avoid data leakage across cohorts and the harmonization of PET data using ComBat are appropriate and well justified. Moreover, the authors addressed limitations in the data structure and validation strategy transparently. The updated spatial analysis with spatial permutation testing and the shift to reporting AMI (Adjusted Mutual Information) rather than ARI further improve the robustness of the findings. In summary, I believe the authors have carefully and convincingly addressed the reviewers' concerns and improved the manuscript substantially. The revisions enhance the clarity, methodological rigor, and clinical relevance of the study. While I have no major concerns remaining, I do have a minor comment for the authors. In the rebuttal, and similarly in manuscript, the authors state: "In the future, when more data becomes available, we plan to incorporate fully quantitative outputs for Aβ and τ." I would appreciate if the authors could further elaborate on this point. Specifically, how much additional data would be required to enable robust quantitative predictions? Given that the present study already combines several large and well-established cohorts, it would be helpful to better understand how realistic this plan is in the context of currently available or foreseeable datasets.

Response R5C1: We sincerely thank the reviewer for their thorough evaluation and positive feedback on our manuscript revisions. We are pleased that our efforts to address the previous round of comments have substantially improved the manuscript's clarity, methodological rigor, and clinical relevance.

Regarding the reviewer's question about our plans to incorporate fully quantitative outputs for Aβ and τ in future work, we appreciate the opportunity to elaborate further. Our current model provides binary classification of Aβ and τ status, which aligns with how these biomarkers are often interpreted clinically. However, moving toward continuous quantitative predictions would provide even greater utility for disease staging and monitoring.

In our current study, we had limited access to raw SUVR maps with harmonized protocols across all cohorts. For robust quantitative prediction of continuous Aβ and τ burden (e.g., global or regional SUVR), accounting for the dimensionality of our feature space and expected effect sizes, we would require a larger number of cases with high-quality, spatially normalized PET images and corresponding multimodal clinical data. Additionally, more diverse cohorts across racial, socioeconomic, and geographic dimensions, with broader representation of comorbidities and mixed pathologies, are required to ensure model validity across populations. Finally, and most importantly, we do believe that integrating the emerging blood-based biomarkers, such as ptau 217, will be key in achieving successful regression tasks.

While currently available multi-site datasets provide a valuable foundation, achieving this scale may depend on ongoing data harmonization efforts (e.g., NACC-SCAN initiatives, or future contributions from clinical trials and observational studies such as AHEAD, Bio-Hermes, or the Framingham Heart Study). We are actively monitoring such efforts and plan to expand our modeling approach as new, large-scale quantitative PET datasets become available. In parallel with these data collection efforts, we are exploring alternative modeling approaches, such as developing ordinal regression models as an intermediate step toward fully continuous predictions.

Comment R5C2: As a critical remark, I do not see that the current approach is really ready to change pathways since blood-based biomarkers alone facilitate AUCs between 0.93 and 0.96 to predict AD neuropathology (i.e., https://www.nature.com/articles/s41591-025-03622-w). The updated AUCs of 0.79-0.84 of the proposed approach look rather low in this comparison. However, I acknowledge the authors idea of integrating blood-based biomarkers in their concept in future.

Response R5C2: We appreciate the reviewer's critical insight regarding the comparison of our approach to emerging blood-based biomarkers, especially in light of recent advances. The reviewer cites the impressive AUCs (0.93-0.96) reported by Ashton et al. in *Nature Medicine* for blood-based biomarkers predicting AD neuropathology. This is an important comparison that deserves clarification. The cited paper primarily uses CSF Aβ42-p-tau181 ratio as its reference for defining Alzheimer’s pathology, specifically leveraging FDA-approved Lumipulse assays and Gaussian mixture modeling to define cutoffs. While a small subset of participants was adjudicated using amyloid-PET, most cases (~95%) relied on CSF-based definitions. This differs substantially from our study, where the reference standard was based on PET imaging of amyloid and tau, which are increasingly being used as endpoints in clinical trials and staging frameworks.

We are certainly aware of how promising plasma p-tau217 is as a biomarker, even when compared directly to amyloid PET. However, it is important to note that performance varies significantly depending on the assays used, the settings in which the biomarker is tested (research cohort versus secondary or primary care), and the type of cut-offs used to define positivity (single- or two-point thresholds). Importantly, different assays capture different aspects of pathology (brain-derived versus peripheral p-tau) and can be affected by non-neurological factors such as chronic kidney disease, body mass index, and cardiovascular health. Nonetheless, we plan to include p-tau217 in our future work once its role in clinical implementation pathways are better established, larger samples of subjects with plasma biomarker data become available, and standardization across different assays is achieved. Several additional considerations help contextualize our work.

Importantly, predicting PET positivity, particularly regional tau stages, from real-world multimodal clinical data (rather than CSF or plasma alone) remains an open and more challenging problem. In addition, our model was intentionally designed to assess tradeoffs between performance and feasibility, by enabling predictions using entire neurological work-up data (e.g., medical history, comorbidities, medications, labs, imaging, cognitive tests) without assuming access to advanced biomarkers such as CSF or plasma. The AUCs and other metrics reported in our study are in that context reflecting a real-world data foundation. That said, we agree with the reviewer that plasma biomarkers have demonstrated excellent diagnostic potential, and our framework is designed to incorporate these inputs once they are clinically available at scale. We believe that integrating routine clinical data with emerging plasma biomarkers will ultimately provide a more generalizable solution for AD staging and prediction, especially across diverse care settings.

Comment R5C3: I only checked the availability and found the code online. I am not an expert in coding, so I cannot judge if it is running properly.

Response R5C3: To support transparency and reproducibility, we have uploaded the complete source code associated with this manuscript to our GitHub repository (https://github.com/vkola-lab/ncomms2025), along with clear documentation and instructions for use. Our lab has a long-standing commitment to open science, and we have made the code for nearly every one of our publications publicly available. These repositories have been widely used by the research community, accumulating hundreds of stars, forks, and pull requests across multiple projects. We aim to make our tools accessible and useful for both technical and non-technical users, and we are happy to provide additional guidance to anyone interested in applying or adapting our models.

Reviewer #6 (Remarks to the Author):

Comment R6C1: I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Response R6C1: We strongly support initiatives like the co-review program that recognize and empower early career researchers. In fact, the first author of this manuscript, Ms. Varuna Jasodanand, recently participated in such a program, and we are grateful to the editorial team for providing meaningful opportunities that foster professional development at this critical stage of scientific training.

Response letter

Reviewer #4:

Comment R4C1: If harmonization and deduplication steps did not matter for the study modeling/conclusions, then that is an important point to show quantitatively in the manuscript.

Response R4C1: Based on the reviewer feedback from previous rounds, we applied ComBat harmonization to reduce variability in tau PET data drawn from multiple cohorts. The revised manuscript includes a description of how harmonization influenced tau PET distribution across regions of interest. Harmonization was important for removing artificial site effects while preserving biological signal (Extended Figure 6; Supplementary tables [20-21]), but we note that core modeling conclusions remained consistent. Please see Supplementary Table 22, which is pasted below, illustrating model performance metrics before and after harmonization.

Regarding deduplication, we did not include duplicated participants at any stage of the analysis to avoid data leakage. To clarify, we originally compared the NACC and OASIS3 samples selected for our study to ensure there were no duplicated subjects. After our first round of revision, in response to reviewer feedback regarding overlaps between OASIS3 and ADNI, we performed similar comparisons between the two and found no duplicated subjects in our original training set.

Label	**AUROC Difference**	**AUPR Difference**
Amyloid	0.03	0.03
Meta temporal tau	-0.02	0
Medial temporal tau	-0.01	0.02
Lateral temporal tau	-0.01	-0.01
Medial parietal tau	-0.03	0.05
Lateral parietal tau	-0.05	0.03
Frontal tau	-0.05	0.04
Occipital tau	-0.03	-0.01

Supplementary Table 22: Performance changes after harmonization of the tau PET data. Values represent the difference in AUROC and AUPR metrics between harmonized and unharmonized models, with positive values indicating improved performance after harmonization.

Comment R4C2: "Ptau217 remains an open issue, and more research is needed before clinical implementation of specific plasma assays" These are widely available for clinical use via CLIA certified laboratories, and one has been FDA approved. Software relying on complex multi-model inputs with unsolved problems in real-world clinical workflows (e.g., combining cognitive assessments, MRI, laboratory tests, co-morbidities and medication data) in a resource poor setting that has a challenging time accurately reporting and organizing this data without also adding an AI modeling layer seems unrealistic, unnecessary, and certainly unproven by the current paper.

Response R4C2: We appreciate the reviewer's emphasis on ensuring realistic and responsible framing. To clarify our study's scope: we do not propose immediate deployment of the model in resource-poor settings, and do not assume full integration of all data modalities. Rather, we present a flexible and modular framework that can make predictions based on the subset of data available for a given individual. This is particularly relevant in workflows where data completeness varies, and multimodal information may be missing. For example, the framework can generate predictions when clinical dementia ratings and cognitive diagnosis are missing.

Regarding p-tau217, we understand that a plasma assay has recently received FDA approval, which marks a significant milestone in the field. question the assay's scientific validity. We recognize initiatives like the Alzheimer's Drug Discovery Foundation's Diagnostics accelerator QC program as important steps towards broader implementation[1]

We have revised our Introduction and Discussion sections to emphasize that this work demonstrates the feasibility of integrating routinely collected clinical data to estimate PET-based biomarkers (amyloid and tau), rather than a ready-to-deploy clinical tool. Specifically, we clarified that our contribution is a research framework for multimodal data fusion to inform future translational efforts.

Comment R4C3: "Our study has a few limitations despite its strengths in scale, multimodal integration, and validation approach. Our model was developed and validated on seven distinct cohorts; however, its generalizability across diverse populations and clinical settings remains to be determined, as the dataset was predominantly composed of White participants. Importantly, due to the lack of non-AD and mixed dementia cases in our datasets, the generalizability of our findings to these important clinical phenotypes remains to be evaluated." This is a good addition, but it should be clear that in resource poor routine clinical settings, non-AD causes of cognitive impairment are quite common making the claims of clinical utility of the current model that is ignorant of this fact overblown.

Response R4C3: We thank the reviewer for highlighting the importance of generalizability, particularly in routine clinical settings where non-AD causes of cognitive impairment are prevalent. We agree that this is a critical issue and appreciate the opportunity to clarify the scope and design of our study.

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[1] https://www.alzdiscovery.org/news-room/announcements/the-alzheimers-drug-discovery-foundations-diagnostics-accelerator-dxa-invests-in-expansion-of-quality-control-program-for-blood-and-csf-biomarkers.

Our model was trained to predict amyloid and tau PET positivity rather than clinical diagnoses such as AD or non-AD dementia. Our goal was to build a flexible, data-driven framework that could estimate PET biomarker status using routinely collected clinical information. The training datasets included individuals with non-AD (n = 969) causes of dementia and our current biomarker prediction model builds upon our previously published framework trained on a larger sample (n = 38,319) for differential diagnosis of dementia, thus exposing it to heterogeneous clinical phenotypes. However, due to limited representation of non-AD and mixed dementia cases in the current PET-labeled testing data, we were unable to examine stratified model performance by dementia subtype in a statistically meaningful way. This limitation has been clarified in the manuscript, and we have acknowledged the need for additional work to assess the model's behavior in more diverse diagnostic subgroups.

In the revised manuscript, we have emphasized that the model is not designed to serve as a diagnostic tool for cognitive impairment nor to differentiate between AD and non-AD etiologies. Rather, it is intended as a scalable and modular approach to estimating amyloid and tau PET profiles, with potential to support downstream applications such as trial screening or research studies. We have also updated the discussion section to more explicitly acknowledge that cognitive impairment in real-world settings often stems from non-AD causes, and that broader validation across these populations is an important area for future work.

> *While our model predicts amyloid and tau PET status as biomarkers of AD pathology, it does not directly distinguish AD from other common causes of cognitive impairment, such as vascular, Lewy body, or mixed dementias. In routine clinical settings, non-AD and mixed etiologies are prevalent, and PET positivity alone may not fully account for the complexity of real-world diagnostic challenges. Therefore, the utility of our approach should be interpreted as a tool for biomarker-based risk stratification, rather than as a comprehensive diagnostic solution for all-cause cognitive impairment.*

Comment R4C4: In general, the clinical applicability and impact of this work is highly speculative, but the manuscript does not read that way. It reads like a major clinical advance, but it is more of a technical proof of potential for multimodal data integration without a sound clinical rational at the moment.

Response R4C4: We thank the reviewer for this important observation. We agree that the clinical applicability of our framework is subject to further validation, and we have revised the manuscript to reflect this more explicitly. Our primary goal is to present a computationally rigorous framework for future translational applications of AI-enabled multimodal integration, and we do not intend to position the work as a major clinical advance at this stage. Throughout the manuscript, we have added revised language to underscore the technical nature of our approach, its dependence on available data infrastructure, and the need for additional research to assess generalizability and clinical translation.

To facilitate future translational research and reproducibility, we have shared our source code within the public domain (https://github.com/vkola-lab/ncomms2025) and developed a prototype implementation of our framework available through Hugging Face (https://huggingface.co/spaces/vkola-lab/ncomms2025). We recognize that this is an initial step, and additional work is needed to translate such tools. In our revision, we further clarified that our work is best viewed as a scalable research tool designed to facilitate future development. We hope the reviewer would find the updated framing to be more reflective of the intent and contribution of our manuscript.

> ***Introduction:*** *By incorporating demographic information, medical history, neuropsychological assessments, genetic markers, neuroimaging and other relevant clinically obtained data, we sought to create a flexible computational framework that explicitly accommodates missing data, reflecting the practical challenges of real-world datasets. Recognizing the synergistic relationship between Aβ and tau pathology in AD pathogenesis, our framework jointly predicts Aβ and τ accumulation to capture their interdependent roles in disease progression. This multi-label prediction strategy addresses key methodological and scientific gaps in existing research, which often considers amyloid or tau in isolation, and serves as a demonstration of scalable participant stratification for research and clinical trials. Finally, by outputting probabilities that align with established biological staging criteria, our modeling framework offers a potential pathway to quantifying disease progression from heterogeneous clinical data.*
>
> ***Discussion:*** *Our approach demonstrates the feasibility of multimodal integration for biomarker prediction, and such frameworks can ultimately contribute to reducing the burden associated with participant selection for AD clinical trials.*

Response letter

Reviewer #4:

Comment R4C1: I appreciate the authors' responsiveness. The methodological advance is clear and well-executed. However, I remain skeptical about its utility for clinical trial stratification or clinical practice. I have no further comments.

Response R4C1: We appreciate the reviewer's recognition of the methodological contributions of our work. We understand the skepticism regarding the immediate clinical utility of such models for trial stratification or routine practice. To clarify, our intention was not to overstate the translational readiness of the framework, but rather to present a technically sound and scalable demonstration of multimodal integration for PET-based biomarker estimation. In our revised manuscript, we explicitly framed the work as proof-of-concept, not a clinical decision-making tool. Specifically, we updated the Introduction and Discussion to highlight that the primary contribution is to lay foundational groundwork for future translational research. We agree that broader validation across diverse, real-world cohorts will be critical to assessing the model's eventual impact in clinical workflows or trial screening pipelines. We hope this clarified framing better reflects the scope and contribution of our study.